

Slope, Parallel Lines and Perpendicular Lines

slopes from coordinates, parallel and perpendicular tests, and classifying figures

Directions

- Slope from two points: $m = \frac{y_2 - y_1}{x_2 - x_1}$ — rise over run, with both differences taken in the *same* order.
- Two lines are **parallel** exactly when their slopes are equal. They are **perpendicular** exactly when the product of their slopes is -1 , that is, when each slope is the negative reciprocal of the other.
- Give every slope as an integer or a fraction in lowest terms, not as a decimal. A vertical line has no slope — write "undefined".

1. Find the slope of the line through $(2, 3)$ and $(8, 7)$.

Answer: _____

2. Find the slope of the line through $(-4, 5)$ and $(6, -1)$.

Answer: _____

3. Find the slope of the line through $(5, -2)$ and $(5, 9)$.

Answer: _____

4. Line j passes through $(-2, 1)$ and $(2, 9)$. Line k passes through $(0, -5)$ and $(3, 1)$.

(a) Find the slope of line j .

Answer: _____

(b) Find the slope of line k .

Answer: _____

(c) State whether the two lines are parallel, perpendicular or neither, and justify your answer using the slopes.

5. Line p passes through $(-1, 4)$ and $(3, 6)$. Line q passes through $(2, -3)$ and $(4, -7)$.

(a) Find the slope of line p .

Answer: _____

(b) Find the slope of line q .

Answer: _____

(c) State whether the two lines are parallel, perpendicular or neither, and justify your answer using the slopes.

6. Find the slope of any line parallel to the line through $(-6, 2)$ and $(2, -4)$.

Answer: _____

7. Find the slope of any line perpendicular to the line through $(3, -5)$ and $(9, 7)$.

Answer: _____

8. Quadrilateral $ABCD$ has vertices $A(-1, 0)$, $B(5, 2)$, $C(7, 8)$ and $D(1, 6)$.

(a) Find the slope of $\overline{AB}$.

Answer: _____

(b) Find the slope of $\overline{DC}$.

Answer: _____

(c) Find the slope of $\overline{AD}$.

Answer: _____

(d) Explain why $ABCD$ is a parallelogram. (You will need the slope of $\overline{BC}$ as well.)

9. Triangle PQR has vertices $P(-2, 1)$, $Q(4, 3)$ and $R(6, -3)$.

(a) Find the slope of $\overline{PQ}$.

Answer: _____

(b) Find the slope of $\overline{QR}$.

Answer: _____

(c) Is the triangle right-angled at Q ? Justify your answer using the slopes.

10. Quadrilateral $EFGH$ has vertices $E(-3, -1)$, $F(1, -3)$, $G(5, 5)$ and $H(1, 7)$.

(a) Find the slope of $\overline{EF}$.

Answer: _____

(b) Find the slope of $\overline{HG}$.

Answer: _____

(c) Find the slope of $\overline{FG}$.

Answer: _____

(d) Using both the parallel and the perpendicular tests, explain what kind of quadrilateral $EFGH$ must be. (You will need the slope of $\overline{EH}$ as well.)